



MANIPAL
UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CYBERSPACE

SUBMITTED BY:

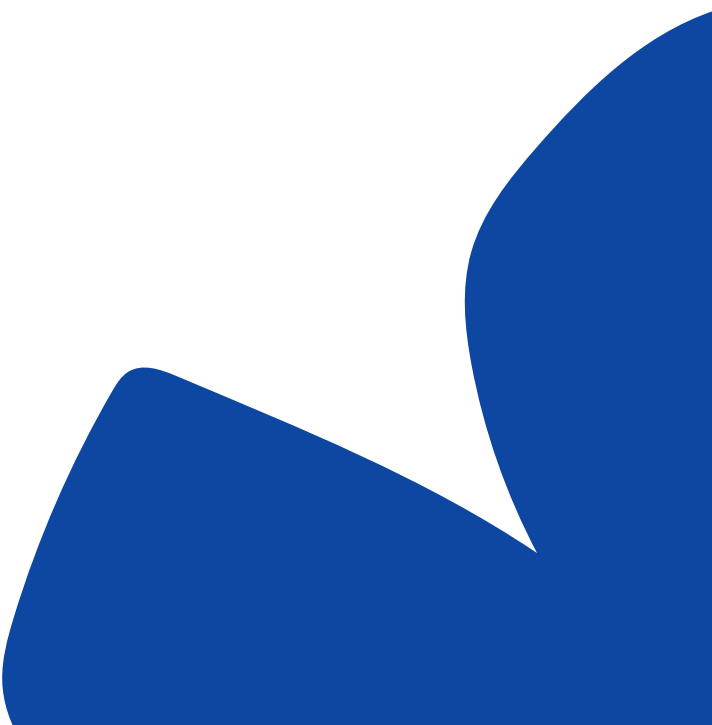
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OUTLINE

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INTRODUCTION

With growing online activity, cyber threats like **phishing**, **fake websites**, and **data breaches** have become common. Many users lack the knowledge or tools to stay protected in this digital environment.

This Project, **CyberSpace** provides a simple and effective solution by offering real-time website checks, email breach detection, and awareness tools to help users browse safely and confidently.



PROBLEM STATEMENT

THE CHALLENGES OF INTERNET SURFING

In today's digital landscape, cyber threats are rising rapidly, targeting individuals and small businesses that lack technical expertise or access to premium cybersecurity tools.

- Millions of users unknowingly visit malicious or phishing websites, risking data theft and financial loss.
- Email breaches expose sensitive personal and organizational information, leading to identity theft and scams.
- Existing cybersecurity solutions are often complex, expensive, and reactive, leaving everyday users vulnerable.

There is a critical need for an accessible, real-time, and educational cybersecurity platform that empowers users to detect threats early, stay safe online, and develop cyber awareness.



PROPOSED SOLUTION

CyberSpace is a SaaS-based cybersecurity platform offering URL scanning, email breach detection, and a browser extension for real-time protection.

Detailed Explanation: Real-time URL analysis using Google Safe Browsing, VirusTotal, PhishTank, etc. Email breach checker integrated with HaveIBeenPwned and breach databases. Cyber Safety Scoring system to educate users and track safety awareness. Chrome extension for seamless browsing protection.

How it Addresses the Problem: Protects everyday users against phishing, malicious sites, and data breaches. Simplifies cybersecurity with one-click scans and reports. Raises cyber awareness with a gamified scoring system. Innovation & Uniqueness: Multi-layered threat detection with educational feedback. Gamified safety score to encourage learning.
Accessible browser extension with privacy-first design



PROJECT OBJECTIVES

- To detect and prevent phishing, malware, and unsafe websites in real time.
 - To identify email breaches and alert users about compromised credentials.
 - To educate users about safe browsing practices through a cyber safety scoring system.
 - To simplify cybersecurity by providing a one-click scanning and reporting platform.
 - To enhance digital awareness and promote responsible online behavior among users.
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TECHNICAL APPROACH

TECHNOLOGIES USED

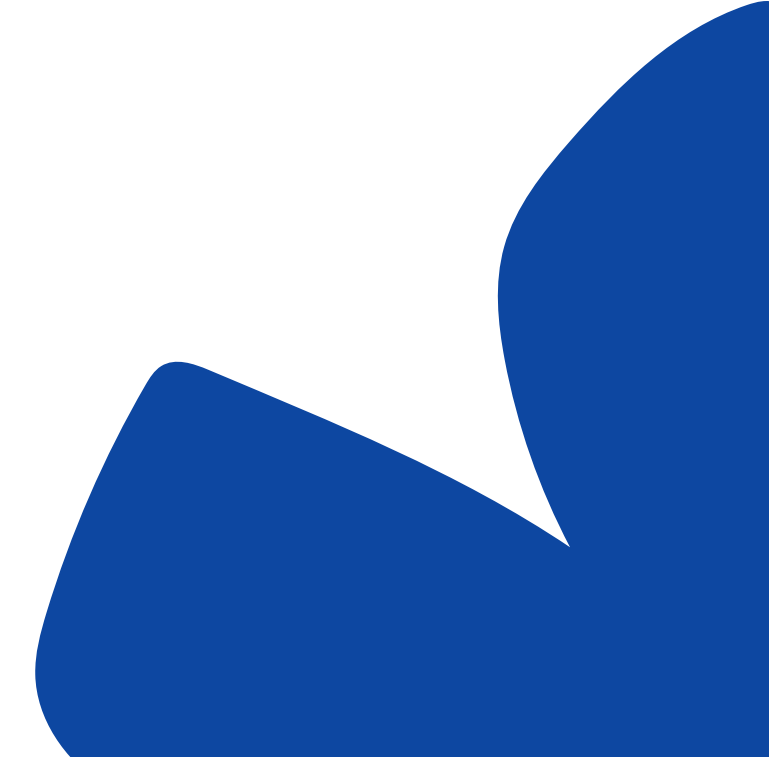
- **Frontend:** HTML5, CSS3, Vanilla JS, Vite
- **Backend:** Node.js, Express.js, Redis (optional caching) Docker-ready
- **Extension:** Chrome Extension (Manifest V3)
- **APIs:** Google Safe Browsing, VirusTotal, HaveIBeenPwned, PhishTank, URLVoid, WHOIS

METHODOLOGY & PROCESS

- User inputs URL/Email
- Backend API aggregates checks from multiple sources
- Risk scoring & analysis generated
- Results shown in web app/extension
- Cyber Safety Score updated and user educated



REFERENCES

- **GOOGLE SAFE BROWSING API → [DEVELOPERS.GOOGLE.COM/SAFE- BROWSING](https://developers.google.com/safe-browsing)**
 - **VIRUSTOTAL API → [DEVELOPERS.VIRUSTOTAL.COM](https://developers.virustotal.com)**
 - **PHISHTANK API → [PHISHTANK.ORG/DEVELOPER_INFO.PHP](https://phishtank.org/developer_info.php)**
 - **URLVOID API → [URLVOID.COM/API](https://urlvoid.com/api)**
 - **WHOIS SERVICES → [WHOISXMLAPI.COM](https://whoisxmlapi.com)**
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THANK YOU